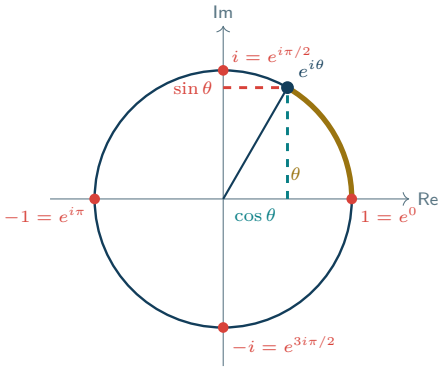
FUNCTIONS LADDER 6 OF 10

Euler's formula and the complex exponential

The exponential of an imaginary number goes round the unit circle, so exponentials and trigonometry are one family.

1 / THE RULE AND THE PICTURE

$e^{i\theta} = \cos\theta + i\sin\theta$ is the point at angle θ on the unit circle



$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

Taking $\theta = \pi$ gives Euler's identity,

$$\boxed{e^{i\pi} + 1 = 0}$$

which links the five constants e, i, π, 1, 0, and a quarter turn $\theta = \frac{\pi}{2}$ gives $e^{i\pi/2} = i$.

2 / WHY IT IS TRUE / TWO DERIVATIONS

Substituting into the series, or following the rotation of the derivative

Substituting ix into $e^x = \sum x^n/n!$ and separating even and odd powers, using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$,

$$e^{i\theta} = \left(1 - \tfrac{\theta^2}{2!} + \tfrac{\theta^4}{4!} - \cdots\right) + i\left(\theta - \tfrac{\theta^3}{3!} + \tfrac{\theta^5}{5!} - \cdots\right) = \cos\theta + i\sin\theta.$$

Geometrically $f(\theta) = e^{i\theta}$ satisfies $f' = if$, and multiplying by i turns a vector by a right angle, so the velocity is always perpendicular to the position and the point moves at unit speed round the circle, covering arc length θ.

3 / SINE AND COSINE AS EXPONENTIALS

Adding and subtracting $e^{i\theta}$ and $e^{-i\theta}$ isolates each

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i},$$

which are the hyperbolic formulas with t replaced by $i\theta$, so $\cos(ix) = \cosh x$ and $\sin(ix) = i\sinh x$; for instance $\cos i = \cosh 1 = 1.5431$ and $\sin i = 1.1752\,i$.
The circular and hyperbolic functions are therefore the same functions seen along the real and imaginary directions.

4 / MULTIPLICATION AND DE MOIVRE / ANGLES ADD

$e^{iA}e^{iB} = e^{i(A+B)}$ contains the addition formulas

Multiplying $(\cos A + i\sin A)(\cos B + i\sin B)$ and comparing real and imaginary parts with $e^{i(A+B)}$ gives $\cos(A+B)$ and $\sin(A+B)$. Repeated multiplication gives de Moivre's theorem $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$, whose real part for $n = 3$ is $\cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta$; at $\theta = 0.7$ both sides equal -0.5048.
The solutions of $z^6 = 1$ are $e^{2\pi ik/6}$, the six vertices of a regular hexagon.

5 / COMPLEX POWERS AND LOGARITHMS

Exponentiation is a growth by e^x and a turn by y

For $z = x + iy$ we have $e^z = e^x(\cos y + i\sin y)$, so $e^{1+i\pi/3} = e(\frac{1}{2} + \frac{\sqrt{3}}{2}i) = 1.359 + 2.354i$.
Logarithms of complex numbers add multiples of $2\pi i$: $\ln(-1) = i\pi$ and $\ln i = \frac{i\pi}{2}$ on the principal branch.
Then $i^i = e^{i\ln i} = e^{-\pi/2} = 0.2079$, a real number, and every other branch gives $e^{-\pi/2 - 2\pi k}$.

ASIDE / ROTATION

Multiplying by $e^{i\theta}$ rotates the plane by θ, which is why alternating current and waves are written as $e^{i\omega t}$.

LINKS TO OTHER SHEETS

The real parts of roots of unity give the chord lengths in *Cyclotomic fields and the arcs of inscribed polygons*; the previous sheet is *Logarithms* and the next is *One equation*, $y'' = \pm y$.

Convention. $i^2 = -1$; angles in radians; $\ln$ on the complex plane is the principal branch with imaginary part in $(-\pi, \pi]$.